

Question	Answer	Marks
2(a)	force per unit mass	B1
2(b)(i)	$g = GM / x^2$	C1
	$= (6.67 \times 10^{-11} \times 1.99 \times 10^{30}) / (1.47 \times 10^{11})^2$ $= 6.14 \times 10^{-3} \text{ N kg}^{-1}$	A1
2(b)(ii)	$E_P = - GMm / x$ $= - (6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.63) / (1.47 \times 10^{11})$	C1
	$= - 2.37 \times 10^9 \text{ J}$	A1
2(c)(i)	$F = L / 4\pi x^2$	C1
	$(g = GM / x^2 \text{ and so } x^2 = GM / g$ and $x^2 = L / 4\pi F$	M1
	elimination of x and subsequent algebra shown leading to $g = 4\pi GMF / L$	A1
2(c)(ii)	correct read-off of pair of values of g and F and full substitution of values of g , G , M and F into equation e.g. $L = (4\pi \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 1.83 \times 10^3) / (8.0 \times 10^{-3})$	C1
	$L = 3.8 \times 10^{26} \text{ W}$	A1
2(c)(iii)	$L = 4\pi\sigma r^2 T^4$ $3.8 \times 10^{26} = (4\pi \times 5.67 \times 10^{-8} \times 5780^4) \times r^2$	C1
	$r = 6.9 \times 10^8 \text{ m}$	A1

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